

Stage 3 — Micro Data

soep-preparation: from raw SOEP to GETTSIM inputs

Where You Are



Same reform. Now on people who exist.

What soep-preparation Tries To Solve

- Everyone is using SOEP data, but every researcher cleans it themselves
 - Setting correct variable data types
 - Giving variables sensible names
 - Recoding negatives to missings or 0
 - Deriving new variables from raw ones
 - Figuring out the reference period of each variable, and aligning them to a common reference period
- In GETTSIM's case also: Make decisions about which SOEP variable corresponds to which GETTSIM input
- soep-preparation is a centralised open-source library that does this for you

What `soep-preparation` Does

Three tiers:

Clean raw variables. Cast each variable to an adequate dtype, convert SOEP missing codes to `pd.NA`, give it a readable and standardized names.

Standard derived variables. Things that follow directly from the raw data — a BMI from height and weight, income from regular employment, etc.

A mapping to GETTSIM's inputs. With an implied recommendation regarding which SOEP variable to use for which GETTSIM input

What It Deliberately Does Not Do

- **Analysis.** It prepares data. It does not run models or produce results.
- **A fixed merged dataset.** You assemble what you need.
- **Reweighting.** SOEP design and longitudinal weights are passed through, not applied.
- **Imputation** beyond what SOEP delivers (exception: assets in 2023).

Cleaning codes and fixing dtypes is the **boring** half of data preparation, and the half that is identical for everyone. The judgement calls (sample selection, which responses are sensible, etc) are still yours to make.

The GETTSIM Mapping and Reference Periods

```
from soep_preparation.config import METADATA
from soep_preparation.gettsim_inputs.mapping import GAP_NOTES, get_soep_to_gettsim

mapping = get_soep_to_gettsim("2023-07-01") # qname -> SOEP variable, or None
mapping["geburtsjahr"] # 'birth_year'

METADATA["gesetzliche_rente_y"]["reference"] # 'previous_year'
```

The reference period (i.e., `current`, `previous_year`, `previous_month` and `time_invariant`) lives in the metadata. See the recipe in the soep-preparation docs (Naming conventions → Reference period).

Testing the Waters

You probably won't have enough time to prepare the SOEP data to get a publishable number.

Don't get stuck on details. Figure out whether the pipeline works for your needs.

Task

10:50–11:55

1. The pytask run has written the cleaned modules and the metadata catalogue to `bld/`.
2. Find the SOEP variables you need using the metadata catalogue and run `create_final_dataset`.
3. Build a mapping table from the GETTSIM inputs you need to the data (`tree` , `dict` or `df_and_mapper`).
4. Fill the inputs you don't have in the SOEP with sensible defaults.
5. Drop households with any unobserved value.
6. Run both environments.